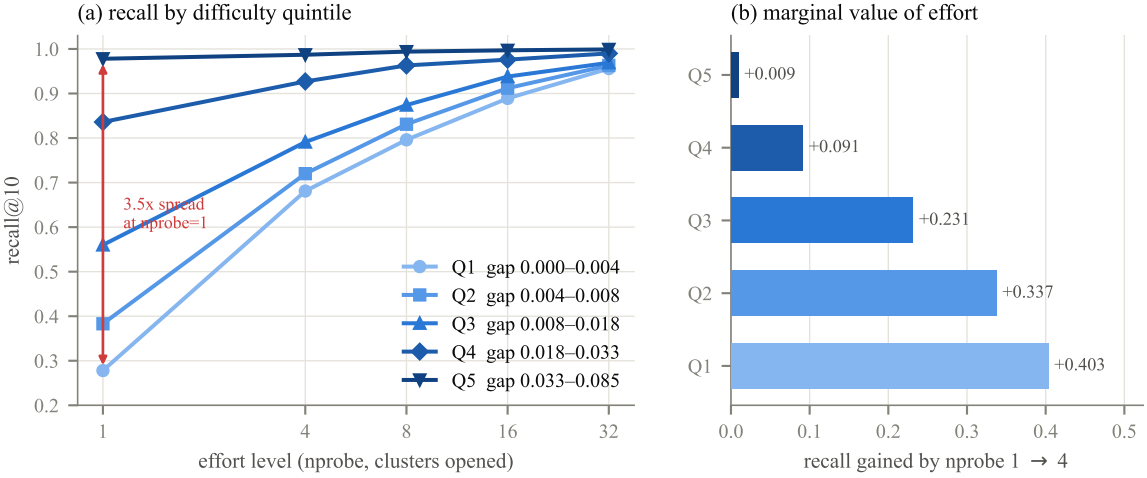
(a) recall by difficulty quintile
recall@10
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
3.5x spread
at nprobe=1
Q1 gap 0.000–0.004
Q2 gap 0.004–0.008
Q3 gap 0.008–0.018
Q4 gap 0.018–0.033
Q5 gap 0.033–0.085
1
4
8
16
32
effort level (nprobe, clusters opened)
(b) marginal value of effort
Q5
Q4
Q3
Q2
Q1
+0.009
+0.091
+0.231
+0.337
+0.403
0.0
0.1
0.2
0.3
0.4
0.5
recall gained by nprobe 1 → 4